PIV Flow-Field Data Analysis and Visualization

English edition · Translated from the Chinese project report

This project analyzes experimental particle image velocimetry (PIV) data using Python to read and process MATLAB-format files. The dataset contains 2,999 instantaneous flow fields, each with 53 × 123 measurement points. Position coordinates and the horizontal and vertical velocity components are recorded at each point. The work covers data loading, visualization, outlier correction, and statistical analysis.

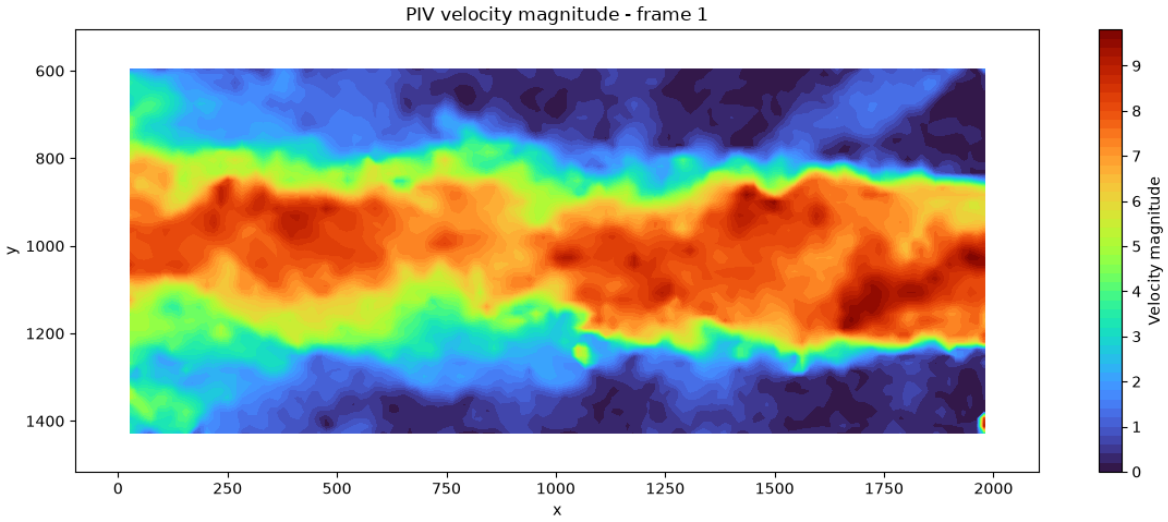
Core tasks

The analysis and visualization are organized into three core tasks. Spatial coordinates x and y are measured in pixels (px); velocity is measured in pixels per frame (px/frame).

Task 1. Flow-field data and visualization

The first task examines an instantaneous flow field from three perspectives: flow direction, velocity magnitude, and spatial profile variation. These visualizations provide a basis for subsequent outlier treatment and statistical analysis. Three types of plots were produced in Python.

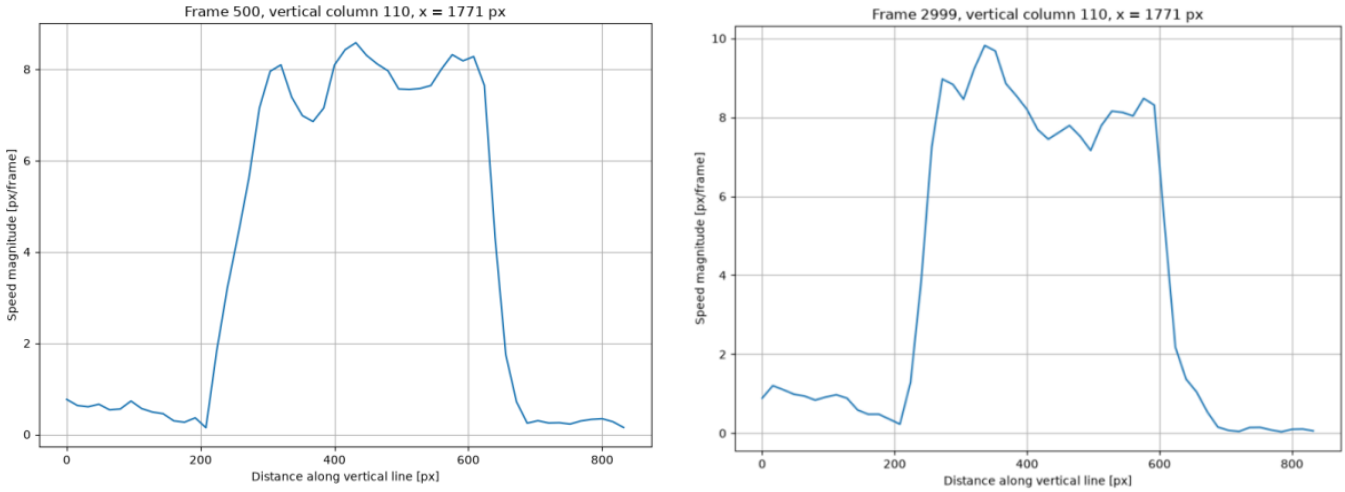
A velocity-magnitude contour plot uses color to distinguish high-speed and low-speed regions and show the spatial distribution of speed across the field.



Core Figure 1. Velocity magnitude in frame 1. Both spatial axes are in pixels. The plot uses 50 color levels, ranging from deep blue for lower speeds to deep red for higher speeds.

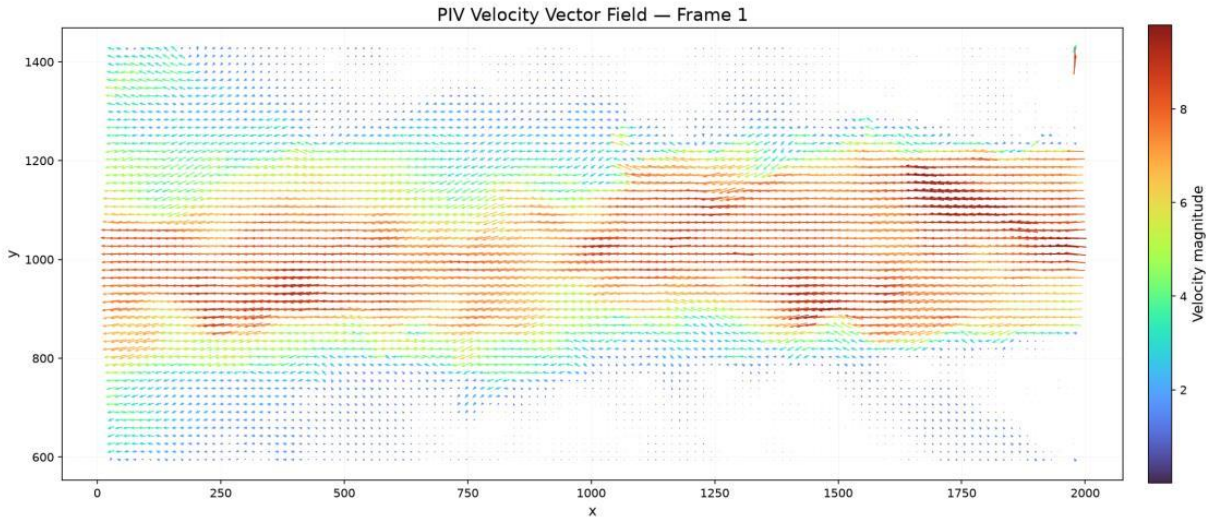
A velocity profile shows how speed changes along a selected spatial line. Profiles help reveal differences between locations and possible local anomalies; numerical checks are still needed to identify outliers reliably. Profiles from different frames were compared at the same measurement column.

Velocity profiles and vector fields



Core Figures 2 and 3. Speed profiles at column 110 (x = 1,771 px) for frames 500 and 2,999. As labeled in the plots, the horizontal axis is distance along the vertical sampling line (px), and the vertical axis is speed (px/frame).

A velocity-vector plot represents the local velocity at each measurement point with an arrow. Arrow orientation indicates flow direction, while color indicates speed: redder arrows correspond to faster flow and bluer arrows to slower flow. This representation helps identify flow patterns and structures and supports later outlier detection and correction. A vector plot was generated for frame 1.



Core Figure 4. Velocity-vector field in frame 1. Although the arrows are densely packed, the overall flow direction and speed distribution remain visible. The color pattern resembles the contour plot, while the arrows additionally show local velocity direction.

Task 2. Detecting and correcting velocity outliers

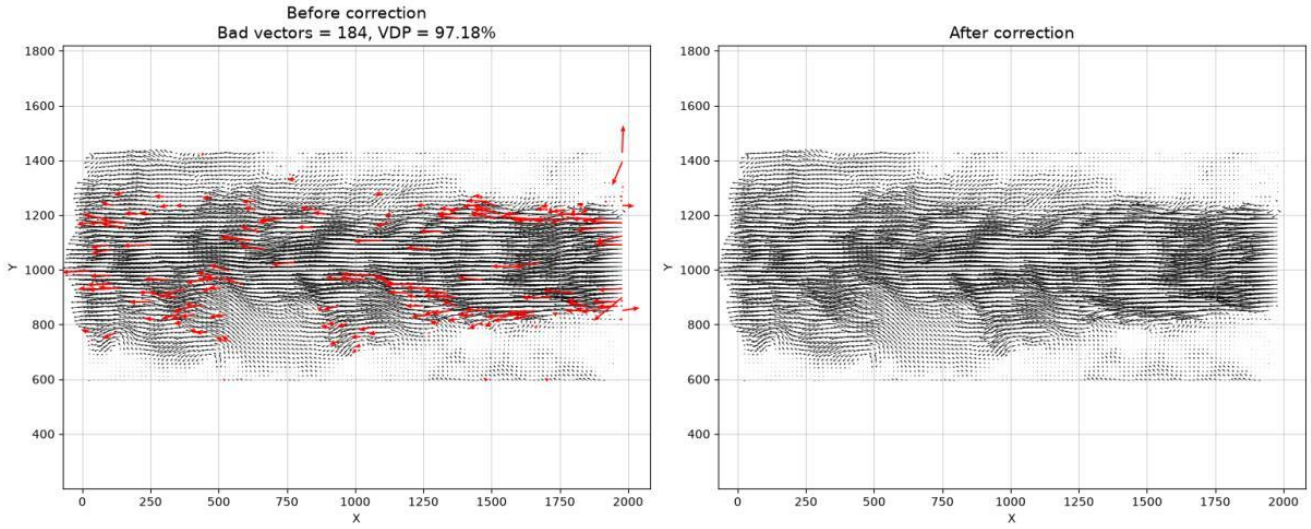
The second task identifies and corrects anomalous velocity vectors in the PIV data to improve data reliability. Inadequate treatment of outliers can affect subsequent calculations of mean velocity and variance. This task requires substantially more programming than the initial visualizations, while using the same dataset and supporting functions.

The procedure uses a 3 × 3 local median filter. At each measurement point, the median of the local neighborhood, including the eight surrounding points, serves as a reference. The deviation of the original velocity from this reference is then evaluated using a three-sigma criterion. A deviation above the mean deviation plus three standard deviations is classified as anomalous and marked in red. Flagged values are replaced by their corresponding local medians to reduce their influence on later statistics.

$$\mu - 3\sigma \leq x \leq \mu + 3\sigma$$

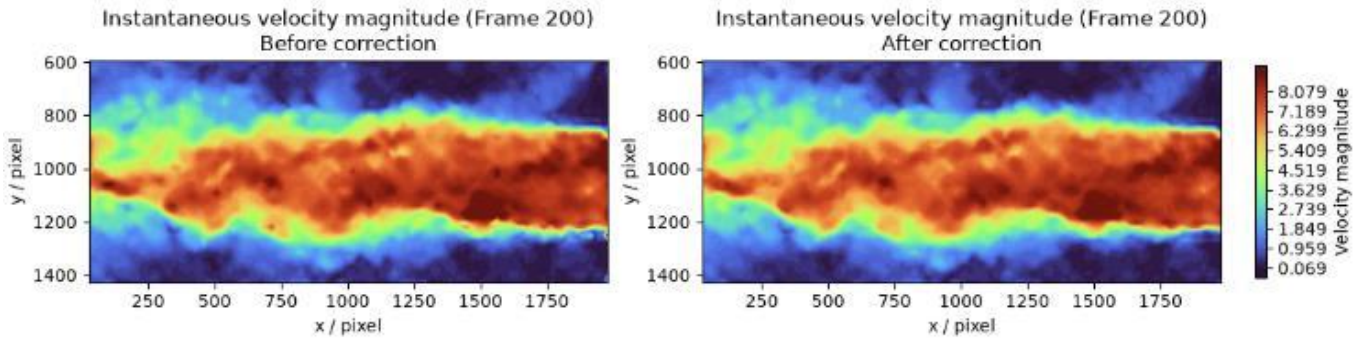
The expression above gives the conventional two-sided three-sigma interval; the deviation-based threshold described here is applied to the deviations from the local median.

Before-and-after vector plots were generated in Python. Normal vectors are shown in black, while detected outliers are shown in red with slightly enlarged arrows. The plots reveal the number and spatial distribution of anomalies in a selected frame.



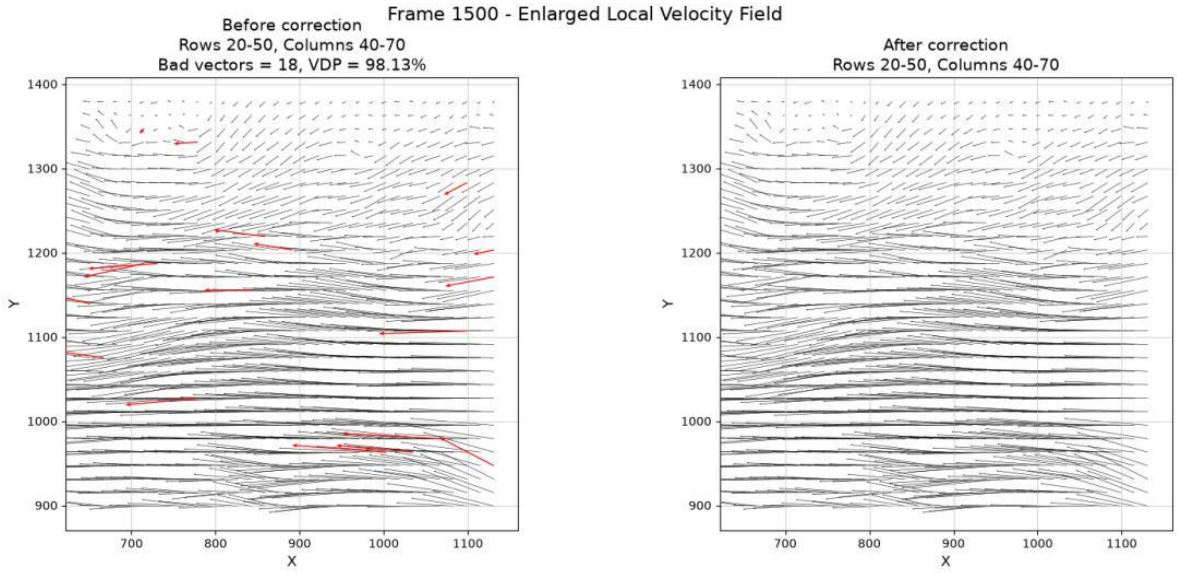
Core Figure 5. Velocity vectors in frame 1,500 before and after correction. Outliers are distributed across the original field. The plot reports 184 anomalous vectors and a valid data percentage (VDP) of 97.18%. Most measurements are therefore retained, although the relatively small anomalous fraction can still affect subsequent calculations.

Visual assessment of outlier correction



Core Figure 6. Instantaneous velocity-magnitude contours for frame 200 before and after outlier correction. The overall pattern changes little because the anomalous fraction is small. The program reports 144 outliers and VDP = 97.79%.

To make individual anomalous vectors easier to inspect, a smaller region was selected from the frame-1,500 data used in Core Figure 5. The enlarged view covers rows 20-50 and columns 40-70, as indicated in the plot labels.



Core Figure 7. Enlarged vector-field comparison for frame 1,500, rows 20-50 and columns 40-70. There are 18 anomalous vectors within this region. The local view makes their removal and replacement much clearer than the full-field comparison.

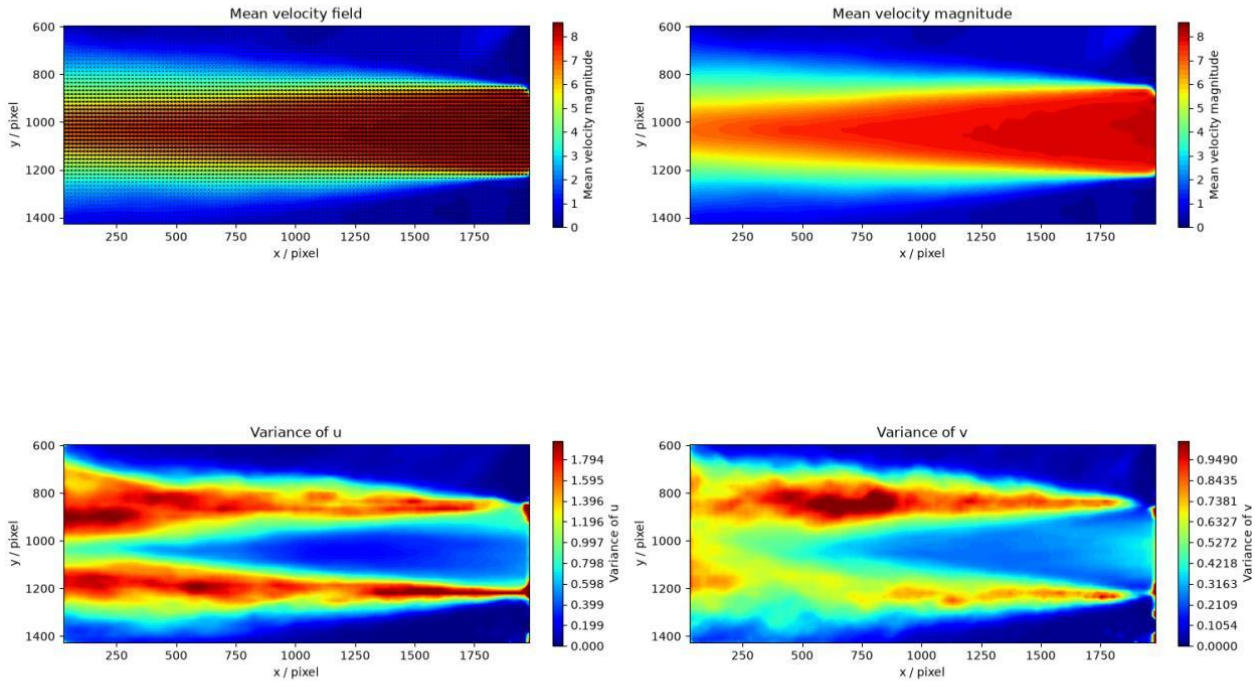
Task 3. Statistical calculation and analysis

The third task calculates statistics, including means and variances, from multiple instantaneous fields. It also examines statistical convergence and the effect of sample size. Relative error measures the percentage difference between a result obtained from the first n frames and the reference result obtained from all 2,999 frames.

$$\text{Relative error} = |V_n - V_{2999}| / |V_{2999}| \times 100\%$$

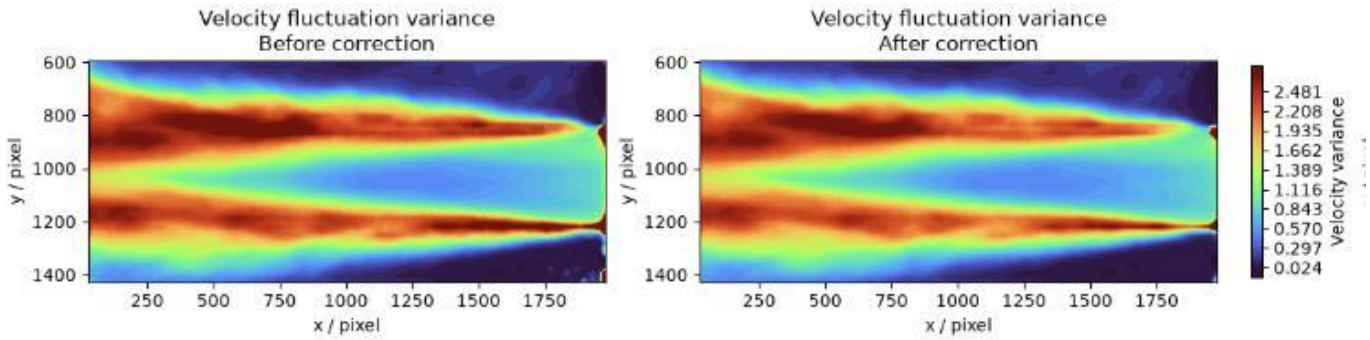
Here V denotes the statistic being compared; the complete 2,999-frame dataset is used as the finite-sample reference.

The data are stacked into a three-dimensional array of shape (2999, 53, 123), and statistics are calculated along the time axis. This produces 6,519 spatial mean values (53 × 123), one at each fixed measurement location. At each location, variance is calculated by averaging the squared differences between the instantaneous velocity and its time mean across the 2,999 frames. Larger variance indicates stronger fluctuations; smaller variance indicates more stable behavior.

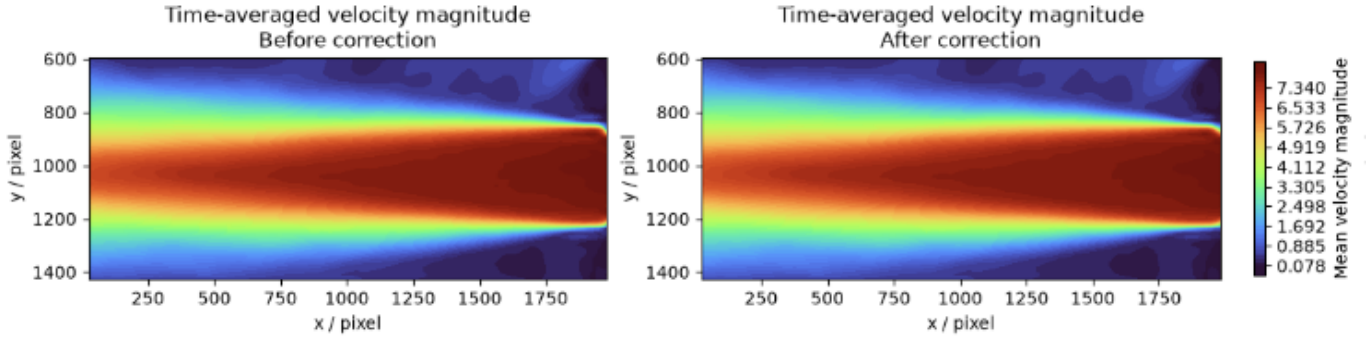


Core Figure 8. Mean velocity field (upper left), mean speed (upper right), variance of u (lower left), and variance of v (lower right). The jet enters from the right. High mean speeds occur in the jet core, while the surrounding fluid moves more slowly. Strong fluctuations occur where the fast jet interacts and mixes with the slower surrounding fluid, especially in the shear layers.

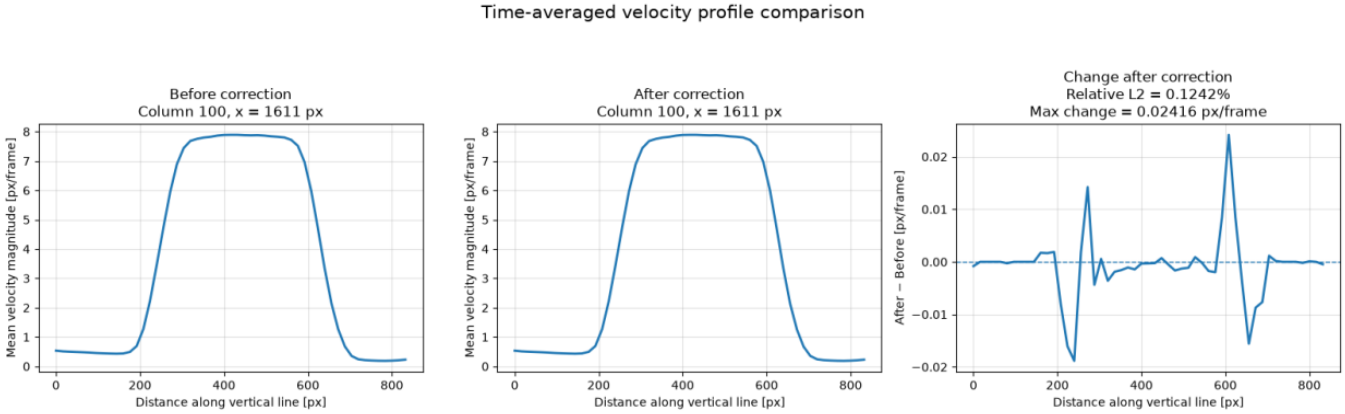
Effect of correction on mean flow and variance



Core Figure 9. Velocity-fluctuation variance before and after outlier correction. Red indicates stronger fluctuations and blue indicates weaker fluctuations. At the full-field scale, the visual differences are small.



Core Figure 10. Time-averaged velocity magnitude before and after correction. The nearly identical large-scale patterns indicate that the detected anomalies have little effect on the overall mean-field visualization.

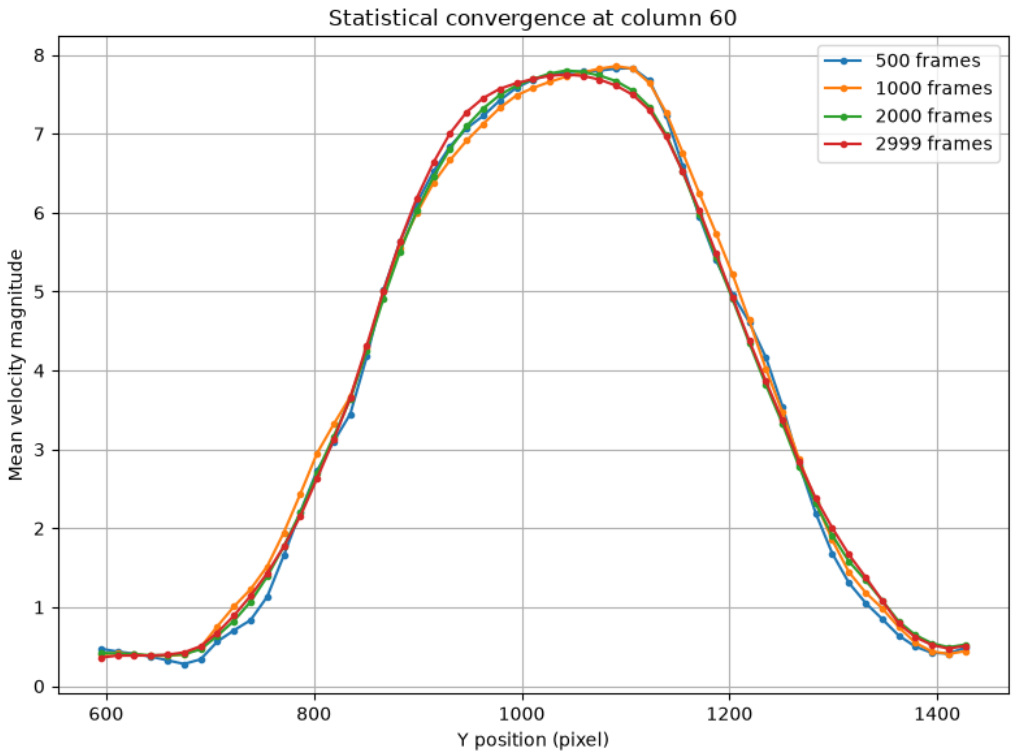


Core Figure 11. Mean-speed profiles at column 100 before and after correction (left and center), with the difference, corrected minus uncorrected, shown on the right. The profiles nearly coincide, indicating that the main flow structure at this location is retained. The reported overall profile difference is about 0.01242%, and the maximum local change is 0.02416 px/frame. Positive differences indicate higher speeds after correction; negative differences indicate lower speeds.

Statistical convergence with increasing sample size

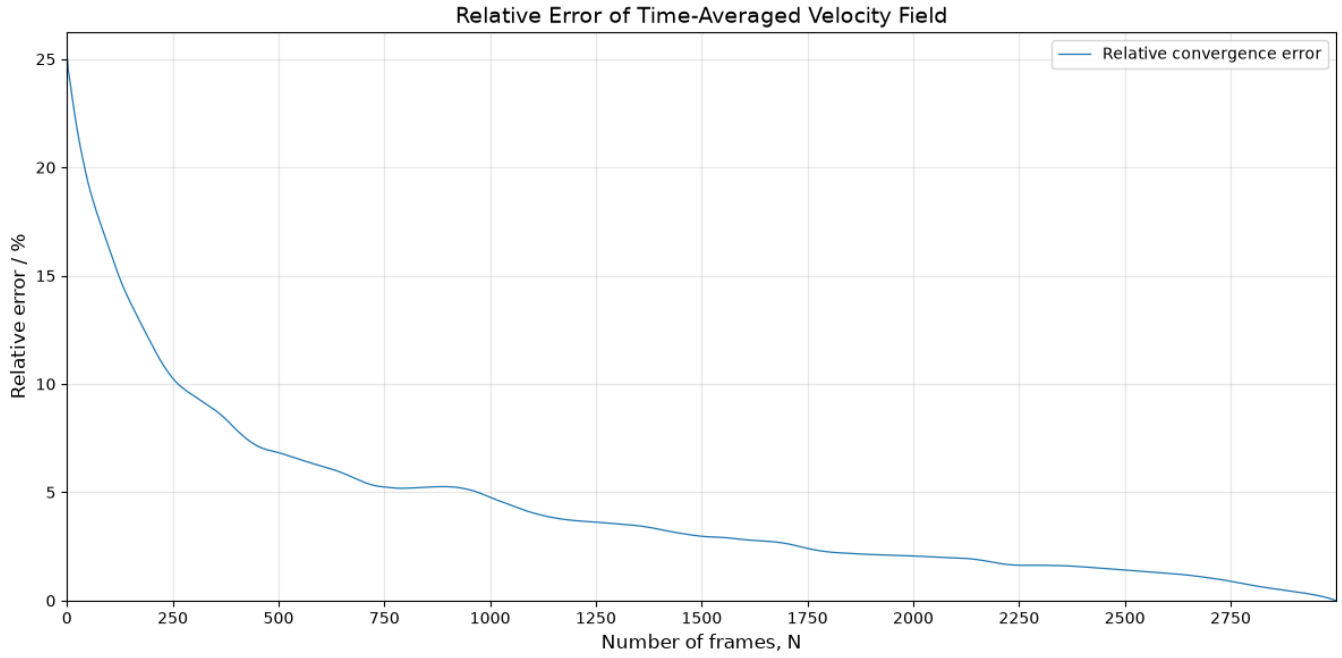
Convergence curves were plotted to assess how the statistics change as more frames are included. A relatively large difference between an early estimate, such as one based on 100 frames, and the 2,999-frame reference is expected. If the discrepancy remains equally large even at 2,000 frames, the data would provide less evidence of convergence. The final 2,999-frame result serves as the comparison reference.

Number of frames	Relative error
500	3.689%
1,000	3.846%
2,000	1.501%
2,999	0.000%



Core Figure 12. Mean-speed profiles at column 60, which contains 53 measurement points, calculated using 500, 1,000, 2,000, and 2,999 frames. The estimates generally approach the full-sample reference as more frames are included. The reported error decreases from 3.689% at 500 frames to 1.501% at 2,000 frames, with a small non-monotonic increase at 1,000 frames.

Convergence across all 2,999 frames



Core Figure 13. Relative error versus the number of frames included in the time-average calculation, from 1 to 2,999. The overall error falls from approximately 25% toward 0%, providing further evidence that the finite-sample estimate stabilizes. The zero at 2,999 frames follows from using that result as the reference.

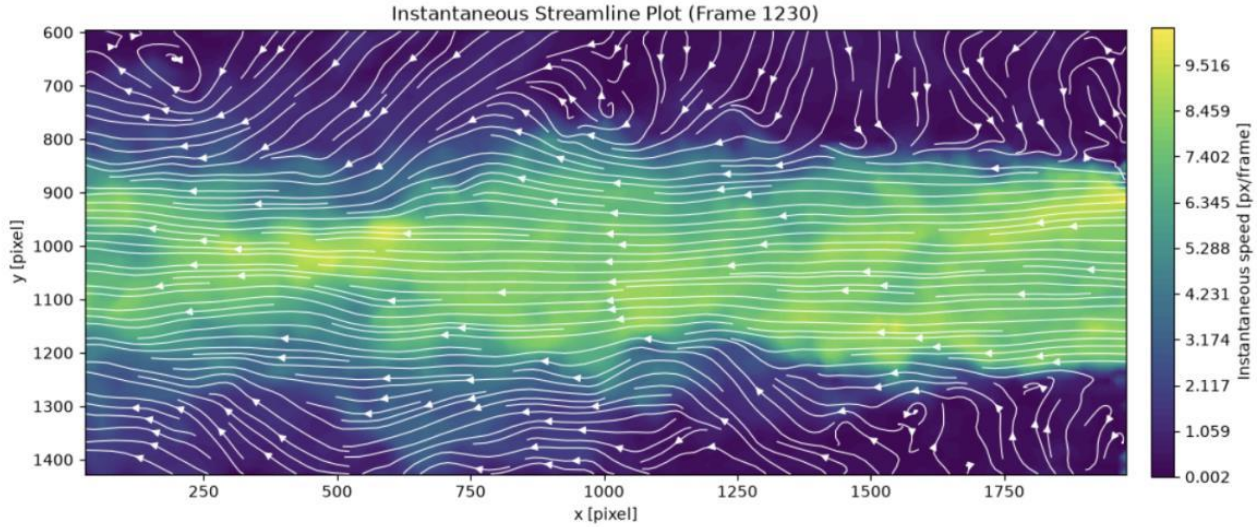
Advanced tasks

Task 1. Streamlines and velocity-profile self-similarity

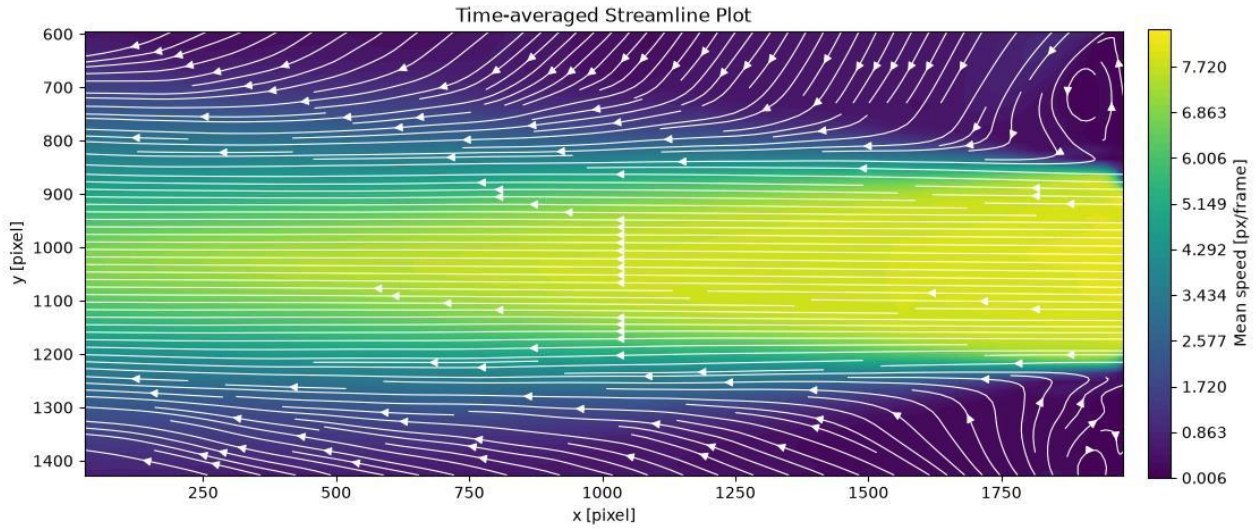
This task produces instantaneous streamline plots, time-averaged streamline plots, original mean-speed profiles, and normalized profiles for self-similarity analysis. An instantaneous streamline plot shows the flow pattern and local direction in a selected frame. White arrows indicate the direction of motion. A time-averaged streamline plot instead represents the mean field obtained from all 2,999 frames.

Self-similarity is assessed by normalizing both velocity and transverse distance. Velocity is scaled by a reference speed, and the transverse coordinate is centered on the jet and scaled by its half-width. This allows profiles with different peak speeds and widths to be compared on the same dimensionless axes.

Instantaneous and time-averaged streamlines

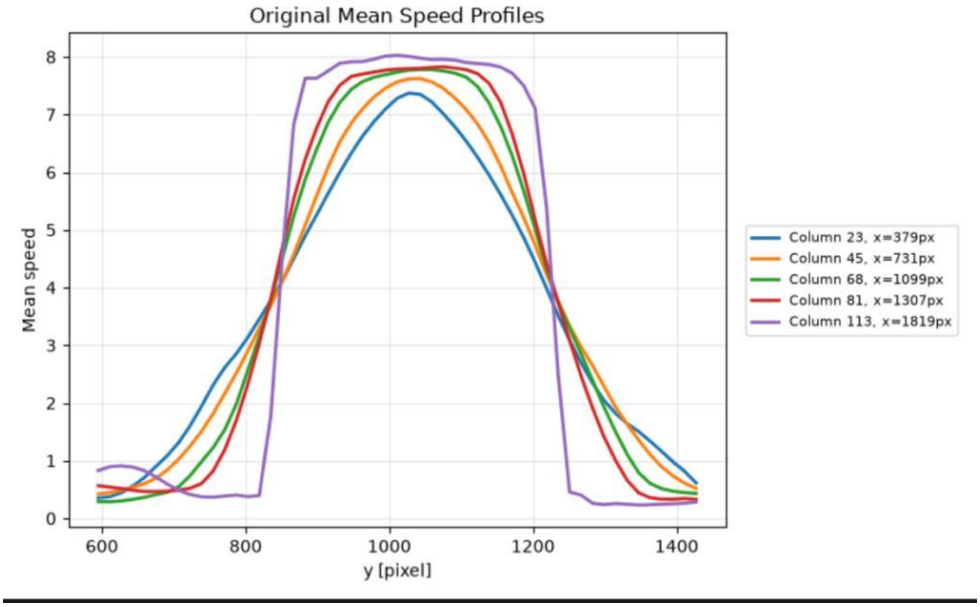


Advanced Figure 1. Instantaneous streamline and speed plot for frame 1,230. Speed is measured in px/frame. Brighter colors indicate faster flow: the yellow-green central region represents the high-speed jet core, while the blue-purple surrounding region has lower speeds. White streamline arrows indicate local flow direction.



Advanced Figure 2. Streamline plot of the time-averaged flow field. The green-yellow central band shows the persistent high-speed jet, and the changing colors toward either side indicate decreasing speed away from the center. White lines and arrows show the mean flow direction. Speed is again measured in px/frame.

Original mean-speed profiles and normalization



Advanced Figure 3. Unnormalized mean-speed profiles at five measurement columns. The horizontal axis is the transverse y position (px); the vertical axis is mean speed (px/frame). The legend gives the measurement column and its streamwise x coordinate.

At column 113 (x = 1,819 px), the peak mean speed is approximately 8 px/frame. Farther downstream, at column 23 (x = 379 px), the peak falls to approximately 7.4 px/frame. Thus, the jet slows as it develops downstream. The upstream purple profile is relatively narrow and steep-sided, whereas the downstream profiles spread over a wider region and have gentler slopes. This behavior reflects mixing with the surrounding fluid and lateral jet spreading.

Self-similarity and velocity normalization

If profiles overlap after each is scaled by its own maximum speed and velocity half-width, they show a degree of self-similarity despite their different dimensional speeds and widths.

$$\text{Normalized speed} = U / U_{max}$$

Dividing by the maximum speed makes the peak equal to 1. A normalized value of 0.5 or 0.8 means that the local speed is 50% or 80% of the profile maximum. This scaling removes differences in peak speed so that profile shapes can be compared.

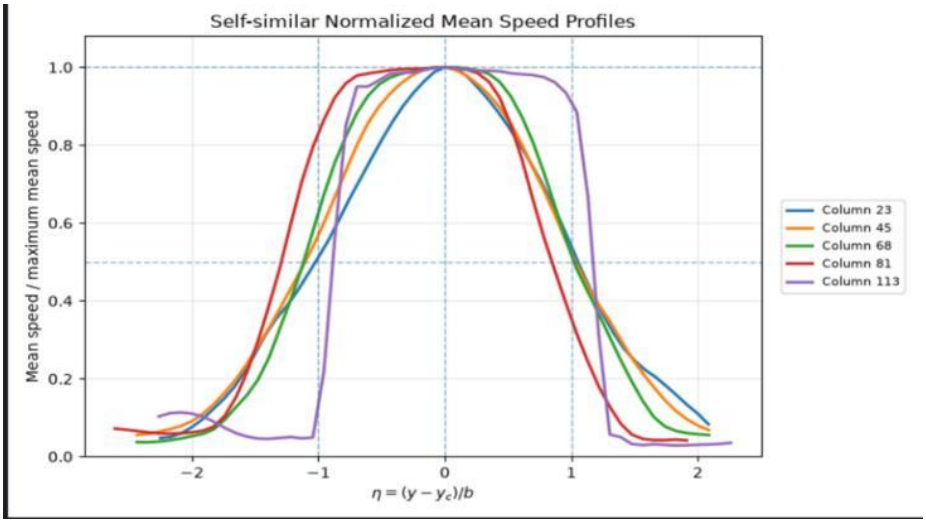
$$\text{Normalized transverse position: } \eta = (y - y_c) / b$$

Here y_c is the jet-center position and b is the half-width. If y_1 and y_2 are the two positions where the speed equals $U_{max}/2$, then $b = (y_2 - y_1)/2$. The normalized coordinate expresses the distance from the center in units of the jet half-width and removes differences in spreading scale.

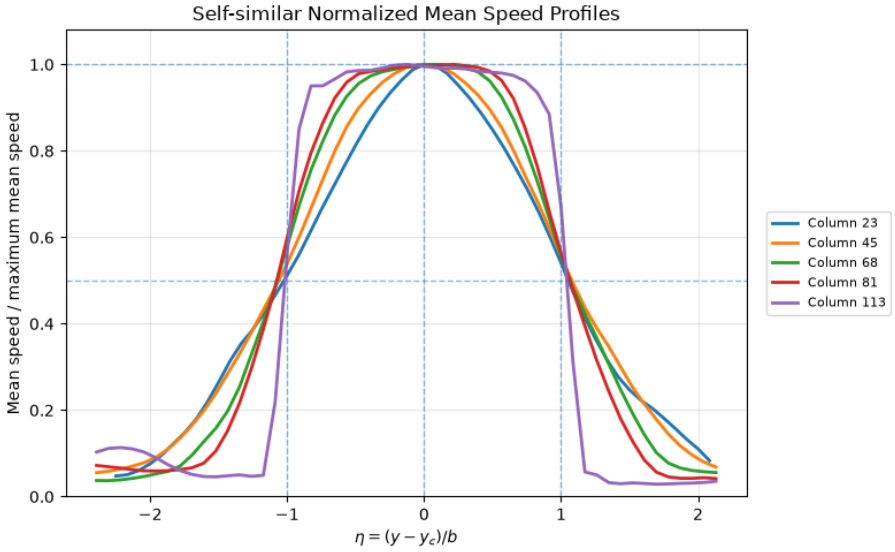
Normalized profiles and sensitivity to half-width

Velocity and width normalization help compare jets at different scales, assess whether the flow is approaching a self-similar developed region, and characterize the shape of downstream profiles.

Half-width estimates can be affected by anomalous measurements or by an unsuitable choice of transverse positions. The two half-maximum crossings should be used consistently: calculate the difference between the larger and smaller crossing positions rather than simply using the maximum coordinate. This reduces the risk that an unintended point controls the width estimate.



Advanced Figure 4. Example of normalized profiles affected by an unreliable width estimate, most visibly for the purple curve at column 113.



Advanced Figure 5. Time-averaged profiles at five columns after velocity and transverse-width normalization. The horizontal axis is $\eta = (y - y_c)/b$, and the vertical axis is U/U_{max}. Partial overlap suggests an emerging trend toward self-similarity as the jet develops downstream.

Task 3. Vorticity, divergence, and confidence intervals

This task explores the concepts of vorticity and divergence and visualizes them with contour plots. It also introduces confidence intervals and bootstrap resampling. **Confidence intervals were not implemented in this project.** The discussion below records the conceptual understanding developed during the work.

1. Vorticity

Vorticity describes the strength and sense of local fluid rotation. In a two-dimensional velocity field, it is the streamwise derivative of the transverse velocity minus the transverse derivative of the streamwise velocity. Its sign indicates rotation direction, while its magnitude indicates the strength of local rotation or shear.

Vorticity is not inherently beneficial or harmful. Strong vorticity can occur in the shear layers between the fast jet and the slower surrounding fluid. A small paddle wheel placed across such a velocity difference would experience unequal forcing on its sides and rotate. Stronger color intensity indicates greater magnitude, while red and blue conventionally denote opposite signs.

As the jet develops, vortical motion in the shear layers draws surrounding slower fluid into the jet. This entrainment contributes to jet widening and a reduction in centerline speed. Because differentiation is sensitive to anomalous velocities, outlier correction should precede vorticity calculation.

2. Divergence

Divergence measures the local tendency of the velocity field to spread outward or converge inward. In the measured plane, it is the sum of the streamwise derivative of the streamwise velocity and the transverse derivative of the transverse velocity. Positive and negative values indicate local outward and inward tendencies, respectively.

For an incompressible flow, a fluid element preserves its volume and the full three-dimensional velocity divergence is zero. This dataset contains only two velocity components in a plane. Its measured in-plane divergence need not vanish unless the omitted out-of-plane contribution also vanishes.

3. Confidence intervals and the bootstrap

A confidence interval uses sample data to estimate a population quantity and expresses uncertainty as a range. In repeated applications of a procedure producing 95% confidence intervals, approximately 95% of those intervals would contain the true quantity. Narrower intervals indicate greater precision; wider intervals can reflect stronger variability, measurement uncertainty, or an insufficient sample size.

Bootstrap resampling estimates uncertainty by repeatedly sampling the observed data with replacement and recalculating a statistic. For example, a resample of 10 observations is drawn from an original set of 10; an observation can be selected more than once, while another may not be selected at all.

Equations and interpretation

Confidence intervals target a population statistic, such as the long-time mean of a stationary jet. Estimates should stabilize with longer sampling. Because the 2,999 frames may be dependent, individual-frame resampling can underestimate uncertainty. A block bootstrap resamples consecutive groups to address temporal dependence; it was discussed but not implemented.

Two-dimensional vorticity

$$\omega = \partial v/\partial x - \partial u/\partial y$$

u and v are the horizontal and vertical velocity components, respectively; x and y are spatial coordinates. The partial derivative $\partial v/\partial x$ measures how v changes with x.

In-plane divergence

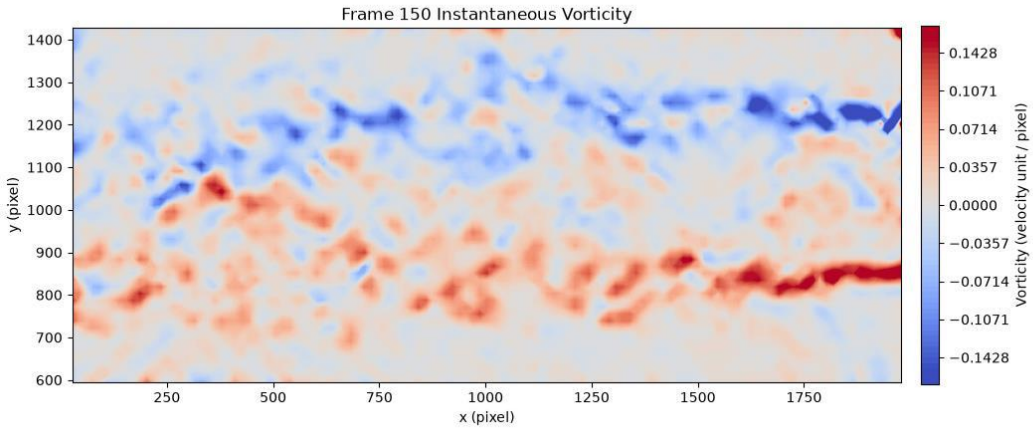
$$\nabla \cdot \mathbf{u} = \partial u/\partial x + \partial v/\partial y$$

The nabla symbol ∇ denotes spatial differentiation. For example, $\partial u/\partial x$ is the rate at which the horizontal velocity changes with the horizontal coordinate.

Conventional confidence interval for a mean

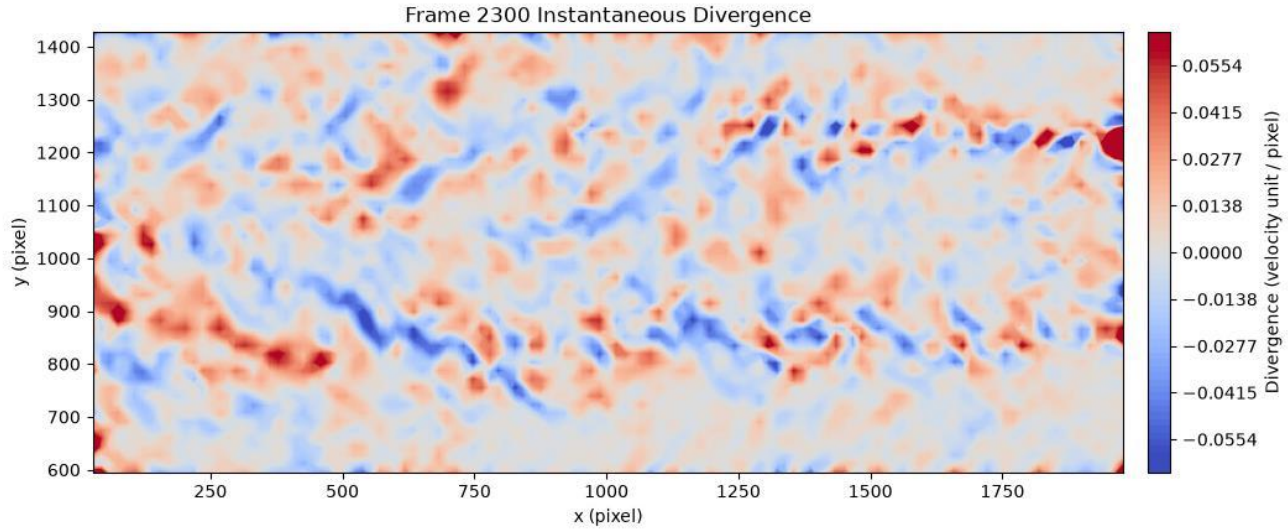
$$95\%\ \mathrm{CI} = \bar{x} \pm t_{0.975,\, n-1} \cdot s / \sqrt{n}$$

CI means confidence interval, $\bar{x}$ is the sample mean, s is the sample standard deviation, and $s/\sqrt{n}$ is the conventional standard error. $t_{0.975,\, n-1}$ is the Student-t quantile. Its use requires appropriate assumptions about the observations; temporal dependence must be considered for this dataset.



Advanced Figure 6. Instantaneous vorticity in frame 150, in frame^{-1}. Red and blue denote positive and negative vorticity, respectively, under the plotted coordinate convention. White indicates values near zero. Stronger colors indicate greater rotation or shear; opposite signs on the two sides of the jet reflect oppositely signed velocity gradients.

In-plane divergence results



Advanced Figure 7. Instantaneous in-plane divergence in frame 2,300, in frame^{-1}. Red indicates positive divergence (an outward tendency), blue indicates negative divergence (an inward tendency), and white indicates values near zero, where the in-plane inflow and outflow approximately balance. Much of the field is lightly colored, indicating relatively small in-plane divergence over much of the measured region.

Scope of the results

The report presents velocity-field visualization, local outlier detection and correction, time-averaged velocity and variance, finite-sample convergence, streamline patterns, profile normalization, vorticity, and in-plane divergence. The confidence-interval and bootstrap discussion is conceptual; no numerical confidence intervals are reported.

English-edition note

The original plots and reported numerical values are retained. Figure numbering is preserved within the core and advanced task sections. Minor wording and notation clarifications align the text with the plotted axes, the displayed row/column selection, and velocity units. These clarifications do not constitute a new analysis of the underlying PIV data.